

Foundations of Natural Language Processing

What is NLP?

Goal: Enable machines to understand human language

Challenges:

- ▶ Ambiguity
- ▶ Context dependence
- ▶ Noise in text

Ambiguity Example

Sentence: "I saw her duck"

Possible meanings:

- ▶ She owns a duck
- ▶ She lowered her head

Insight: Language understanding requires context

Levels of Language

- ▶ Phonology
- ▶ Morphology
- ▶ Syntax
- ▶ Semantics
- ▶ Pragmatics

NLP attempts to model these computationally

NLP Pipeline

1. Raw Text
2. Preprocessing
3. Representation
4. Modeling
5. Evaluation

Text Preprocessing: Motivation

Raw text is messy

Example: "OMG!!! NLP is AMAZING :)"

Goal: Convert into structured format

Case Folding

Idea: Convert text to lowercase

Example: HELLO $\rightarrow$ hello

Why? Avoid duplicate representations

Tokenization

Idea: Split text into words

Example: "NLP is fun" → [NLP, is, fun]

Challenges: punctuation, emojis, contractions

Stopword Removal

Idea: Remove common words

Example: "This is a good book" → [good, book]

Caution: May remove useful information

Stemming

Idea: Rule-based truncation

Examples: running → runn
playing → play

Limitation: Not linguistically accurate

Lemmatization

Idea: Use vocabulary and grammar

Examples: running → run

better → good

Trade-off: Slower but more accurate

Transition

Once text is cleaned...
How do we compare strings?

String Similarity

Goal: Measure similarity between strings

Applications:

- ▶ Spell correction
- ▶ Search engines
- ▶ Plagiarism detection

Edit Distance: Intuition

Idea: Minimum edits to convert one string to another

Operations:

- ▶ Insert
- ▶ Delete
- ▶ Replace

Example

kitten $\rightarrow$ sitting

k $\rightarrow$ s

e $\rightarrow$ i

insert g

Distance = 3

Dynamic Programming Idea

Insight: Reuse subproblems

Build matrix of size $m \times n$

Levenshtein Recurrence

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 \\ D(i, j-1) + 1 \\ D(i-1, j-1) + 1 \end{cases}$$

Interpretation

- ▶ Lower distance $\rightarrow$ more similar
- ▶ Useful for typo correction

Deterministic methods are limited
Let's use probability

Noisy Channel Model

Problem: Observed word x , true word w

Goal:

$$\arg \max_w P(w|x)$$

Bayes Rule

$$P(w|x) = \frac{P(x|w)P(w)}{P(x)}$$

$$\Rightarrow \arg \max_w P(x|w)P(w)$$

Interpretation

$P(w)$: Language model

$P(x|w)$: Error model

Combine likelihood + prior

Example

Observed: speling

Candidates:

- ▶ spelling
- ▶ spilling

Choose max:

$$P(x|w)P(w)$$

Transition

Now extend from one word to full sentence

Sentence Correction Problem

Observed: I hav a dreem

Goal: Find most likely correct sentence

Why Not Independent Correction?

hav → has

Result: I has a dream

Insight: Words depend on context

Hidden Markov Model

Hidden states: Correct words

Observations: Misspelled words

Transition:

$$P(w_i | w_{i-1})$$

Emission:

$$P(x_i | w_i)$$

Viterbi Algorithm

Goal: Find best sentence

Approach: Dynamic Programming

Keep best sequence at each step

Viterbi Recurrence

$$\delta_t(j) = \max_i [\delta_{t-1}(i)P(w_j|w_i)]P(x_t|w_j)$$

Key Insight

- ▶ Avoid trying all sentences
- ▶ Keep only best path
- ▶ Efficient decoding

Wrap Up

- ▶ Preprocessing → clean text
- ▶ Edit distance → similarity
- ▶ Noisy channel → probabilistic correction
- ▶ HMM + Viterbi → sentence correction